

Data, Computation & Innovation II

Instructor: Jared Whalen

Syllabus Stuff

Course Overview

Week 1 (June 3): Chart Fundamentals + SVG/Svelte Introduction

Week 2 (June 10): D3 Fundamentals, Part 1

Week 3 (June 17): D3 Fundamentals, Part 2

Week 4 (June 24): Interactivity and animation + *Guest speaker Kavya Beheraj/The Athletic*

Week 5 (July 1): GIS and Mapping + *Guest speaker Alvin Chang/The Pudding*

Week 6 (July 8): Scrollytelling + Advanced patterns

Week 7 (July 15): Final project presentations

Goals

The goal of this course is to

- become familiar with the tooling for JavaScript-based visualization development
- and to leave knowing enough to keep learning in whatever direction your work takes you.

Expectations

You will (*probably*) not become a JavaScript expert, but you'll (*hopefully*)

- understand enough to build real things
- adapt other people's code
- recognize what you don't know

Tools

Tools we'll be using

- HTML/CSS/JS (the basics)
- Svelte (a UI framework used by many newsrooms)
- D3 (a JavaScript library for bespoke data visualization — but is not a charting library)
- GitHub
- VS Code (or similar)

Deliverables

You will make...

- Charts! (every week) and some maps
- A portfolio-worthy interactive data visualization

This is a making class. Most of what you learn will come from building things, breaking them, and building them again.

Assignments

Weekly assignments

- Those aforementioned charts
 - Expect multiple chart assignments each week between class.
 - They are graded on completion.
 - The goal is repetition and iteration.
- Contributing to the weekly chart discussion thread on CourseWorks
- A final project update (starting Week 3)

Assignments

Final project

- The final project is a portfolio-worthy interactive data visualization.
- The project should be able to stand on its own. That might mean a data story with charts woven into a narrative, a visual essay where the visuals carry the argument, or a dashboard with enough framing and annotation to guide the reader without requiring explanation.
- Week 7 is dedicated to final presentations.

In-class structure

Every week, you can expect...

- Opening critique of peer work in small groups
- 2–3 learning segments
- Open lab period for building/feedback

Critique

- Most classes will open with small-group critique of the previous week's work.
- The format is simple: what's working, what's unclear, ask specific questions, give specific suggestions.
- Learning to edit is one of the most valuable skills you can develop.
- That process can be uncomfortable. But the discomfort is part of it. Come prepared to give feedback that's honest and specific, and to receive it the same way.

On Using AI

- AI tools are part of the industry and ignoring them would be a disservice.
- But this course exists to teach you how things work.
- The expected use of AI in this course is as a learning aid: to explain code you don't understand, to help debug a specific error, or to explore an approach before implementing it yourself.
- Disclose where and how you use AI. This is a skill in and of itself and will help you think through problems.
- Submitting AI-generated or copied work without acknowledgment is a violation of the school's code of conduct.

Questions?

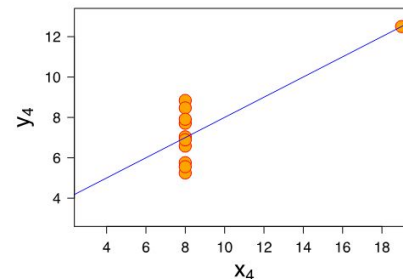
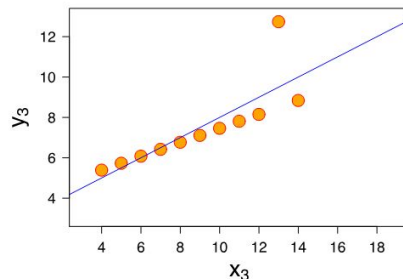
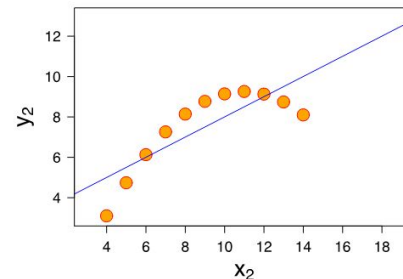
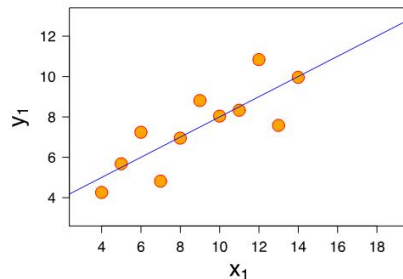
About the course, myself, newsrooms, etc.

Chart Fundamentals

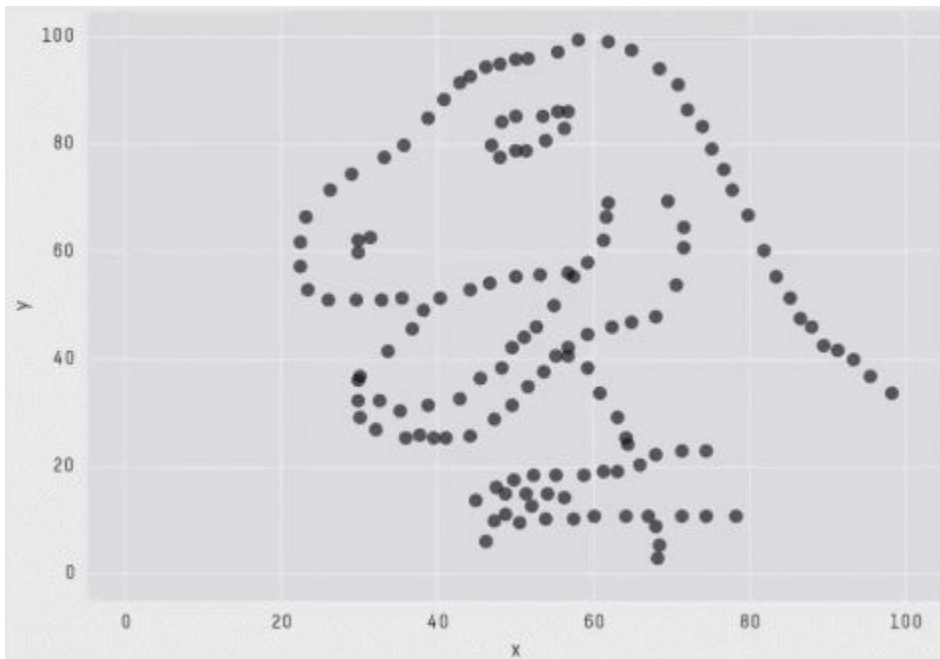
Why we visualize

You have a dataset. You run your summary functions: mean, variance, correlation. Based on these values, the data appears identical. But each has a very different distribution.

This is why we visualize.



Anscombe's quartet, [Wikipedia](https://en.wikipedia.org/wiki/Anscombe%27s_quartet)



X Mean: 54.2659224
Y Mean: 47.8313999
X SD : 16.7649829
Y SD : 26.9342120
Corr. : -0.0642526

Datasaurus Dozen, Revolutionary Analytics

So then what is a chart?

A chart is a way of making patterns in a table visible. That's it. Before any tools, any code, any design, it's a translation from rows and columns into something the eye can process faster than the brain can read numbers.

How often U.K. university students said they use AI programs or feature

	A	B	C
1	Frequency	Female	Male
2	Every day	6	10
3	Every few days	9	18
4	About once a week	12	15
5	Once or twice a month	15	15
6	Less often than once a month	28	21
7	Never	26	18

Source: [YouGov](#) via [Datawrapper](#)

The AI gender imbalance

How often U.K. university students said they use AI programs or features when studying:

Female Male

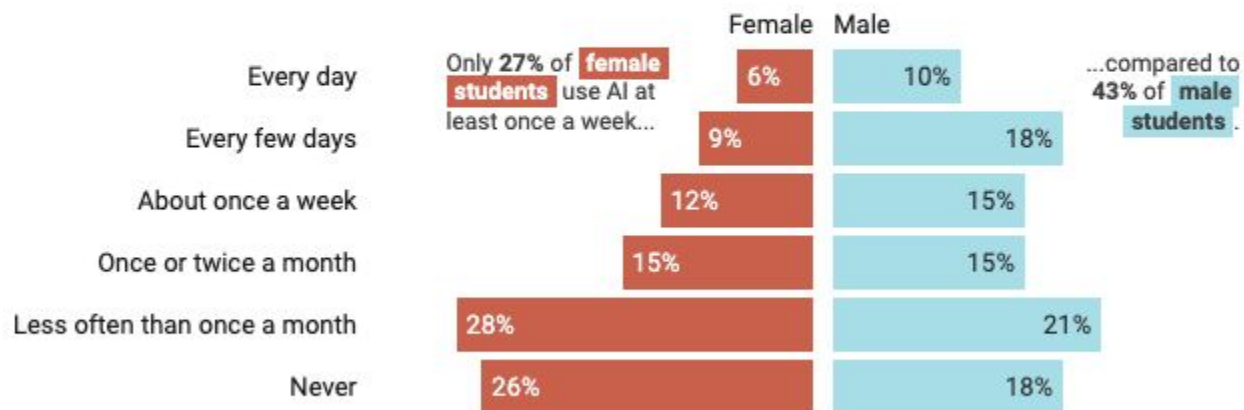


Chart: Elliot Bentley • Source: [YouGov](#) • [Get the data](#)

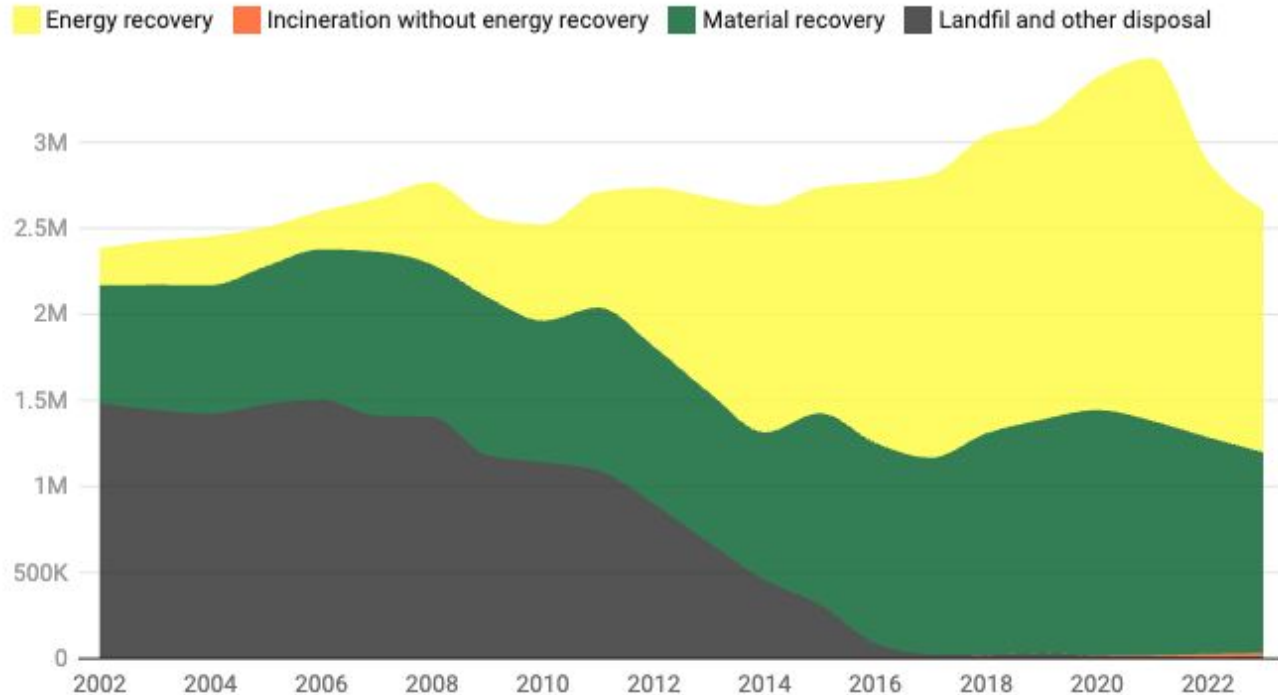
Source: [YouGov](#) via [Datawrapper](#)

Municipal waste by treatment method in Finland, 2018-2024

	A	B	C	D	E
1	Year	Energy recovery	Incineration without energy recovery	Material recovery	Landfil and other disposal
2	2002	216314	0	683543	1484559
3	2003	256311	0	726671	1444541
4	2004	284895	0	745852	1422692
5	2005	226843	0	800784	1478123
6	2006	222323	0	873878	1503910
7	2007	310350	0	953274	1411294
8	2008	477794	0	884304	1406105
9	2009	462750	0	919830	1179840
10	2010	557131	0	822152	1140721
11	2011	672209	0	946234	1093277
12	2012	924825	0	912596	900674
13	2013	1136939	0	872210	672398
14	2014	1315820	0	856326	457733
15	2015	1312180	0	1111338	314762
16	2016	1514830	0	1163566	89535
17	2017	1645554	0	1140133	25902
18	2018	1732450	972	1285711	21949
19	2019	1735109	700	1357227	29669
20	2020	1931974	2522	1424280	18016
21	2021	2109565	6088	1360524	14499
22	2022	1604394	16947	1257747	10660
23	2023	1409696	20215	1165355	12634

Source: [Statistics Finland](#) via [Datawrapper](#)

Landfills are replaced with energy and material recovery



↑ Hover over the areas or legend to see details. ↑

Chart: Linus Aarnio • Source: [Statistics Finland](#) • [Get the data](#)

Source: [Statistics Finland](#) via [Datawrapper](#)

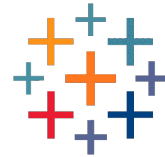
Why not just use...

Datawrapper

***Flourish**



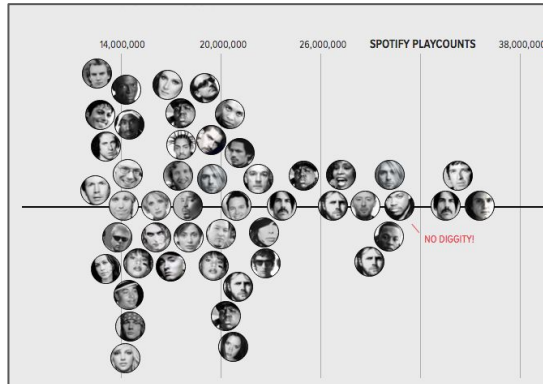
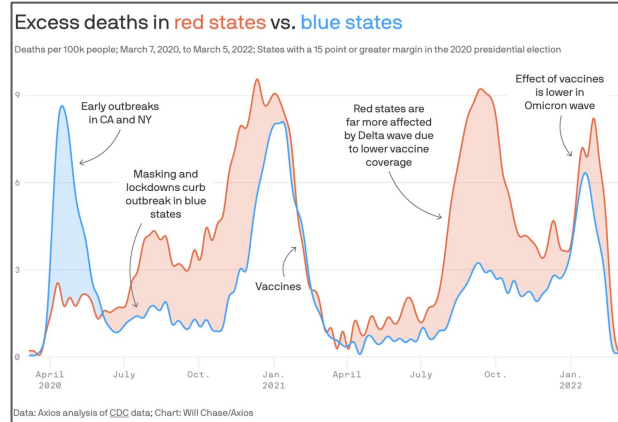
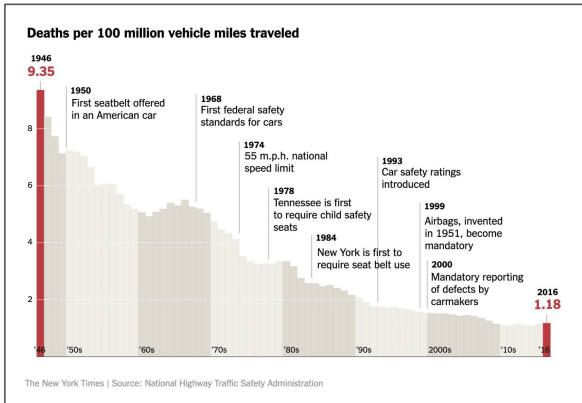
HIGHCHARTS®



+ a b l e a u

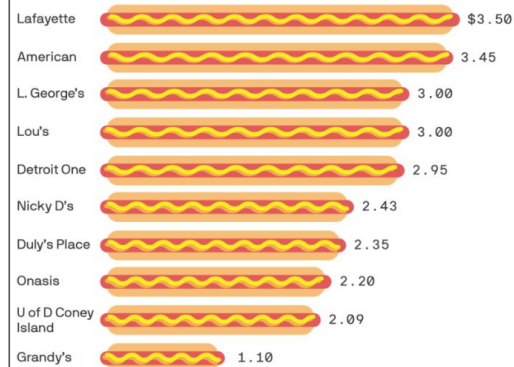
The black box problem





Price of a coney dog at select Detroit restaurants

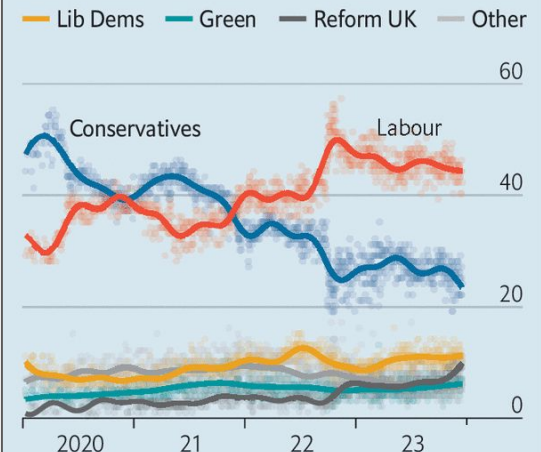
As of April 26, 2022



Date: Axios research; Chart: Erin Davis/Axios Visuals; Illustration: Gabriella Turrisi

Tories in trouble

Britain, general-election voting intention, %



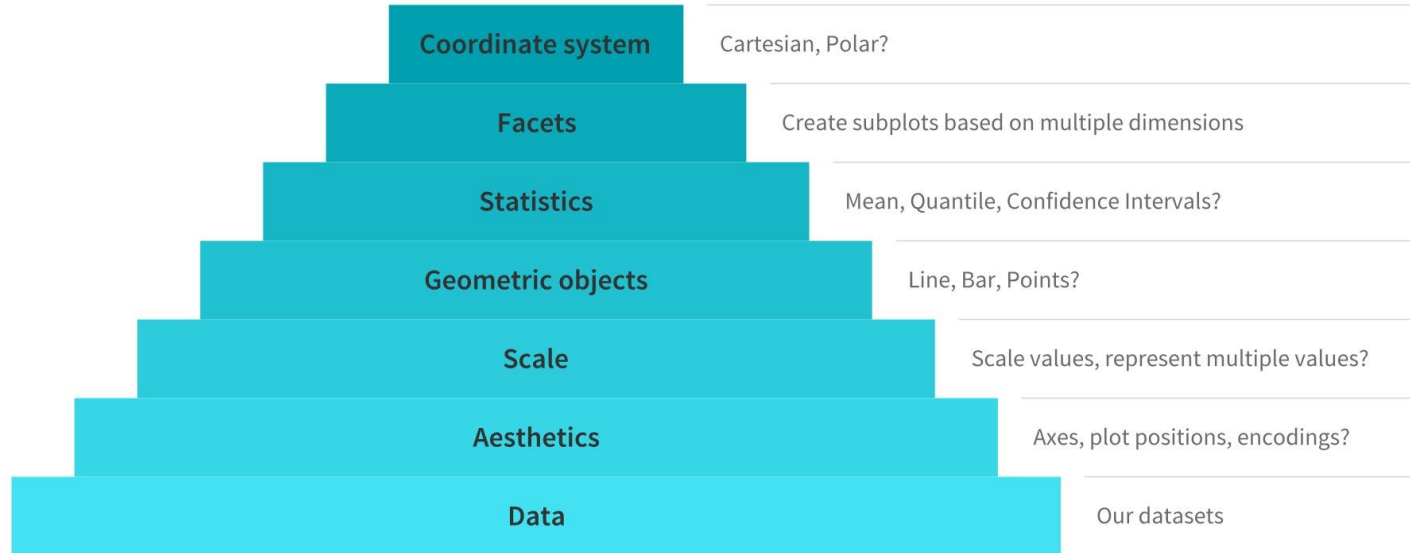
Source: The Economist's UK poll tracker

The Economist

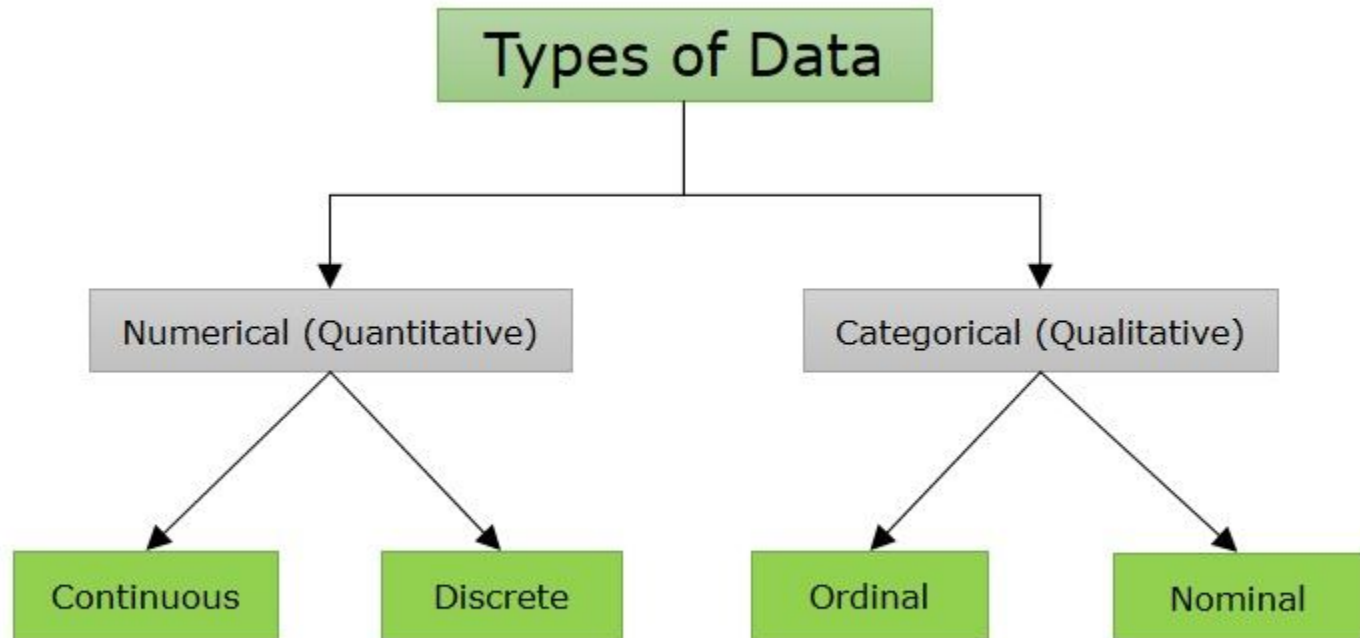
The building blocks

- **Data:** the table everything comes from. Rows are observations, columns are attributes.
- **Geometric objects (marks):** the shapes that represent your data. Points, lines, bars, areas.
- **Aesthetics (channels):** the visual properties of those marks that encode data. Position, length, color, size, shape.
- **Scales:** the translation between a data value and a visual value.

Major Components of the Grammar of Graphics



Source: Saketh Ram Gangam, via [Medium](#)



Numerical (Quantitative) Data

Data that can be measured, counted or given a numerical value.

- Discrete Data

- The discrete data is numerical data that is countable, finite, and can only take certain values, usually whole numbers.

- Continuous Data

- The continuous data is numerical data that can take any value within a specified range including fractions and decimals.

Categorical (Qualitative) Data

Data that can be categorized with or without a meaningful order.

- **Nominal Data**
 - The nominal data is categorical data that can not be arranged in an order or rank.
- **Ordinal Data**
 - The ordinal data is categorical data can be ordered or ranked with a specific attribute.

Geometric Objects (Marks)

The shapes that represent your data.

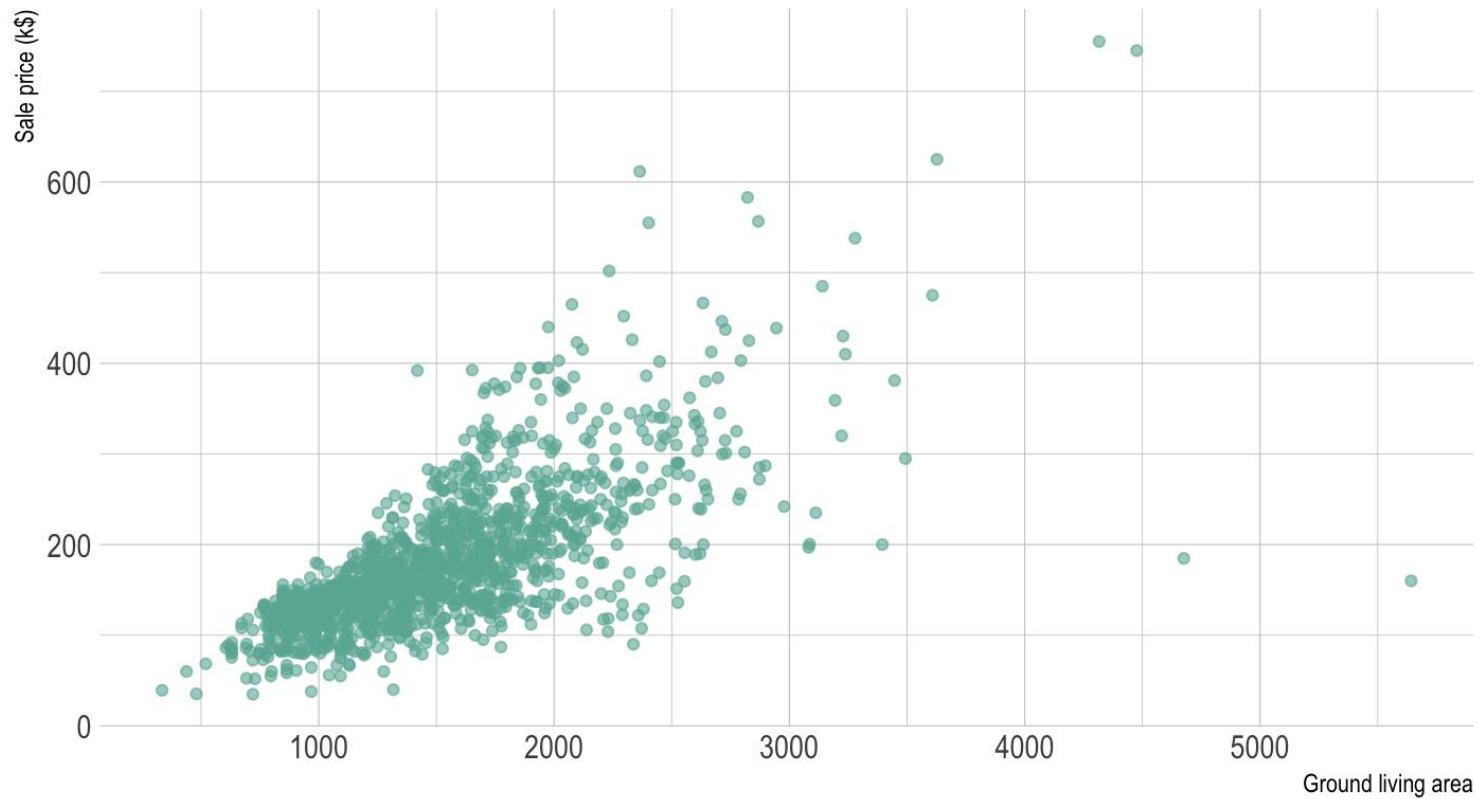
- **Point:** one dot per row. Scatterplots, dot plots, maps.
- **Line:** connects points to show change or sequence.
- **Bar:** a rectangle extending from a baseline to a value.
- **Area:** a filled region, often showing cumulative or proportional values.

Aesthetics (Channels)

The visual properties of a mark that carry information. When you change an aesthetic, you're encoding something. Not to be confused with themes styles.

- Position (x, y)
- Length
- Color (hue)
- Color (intensity)
- Size/Area
- Shape

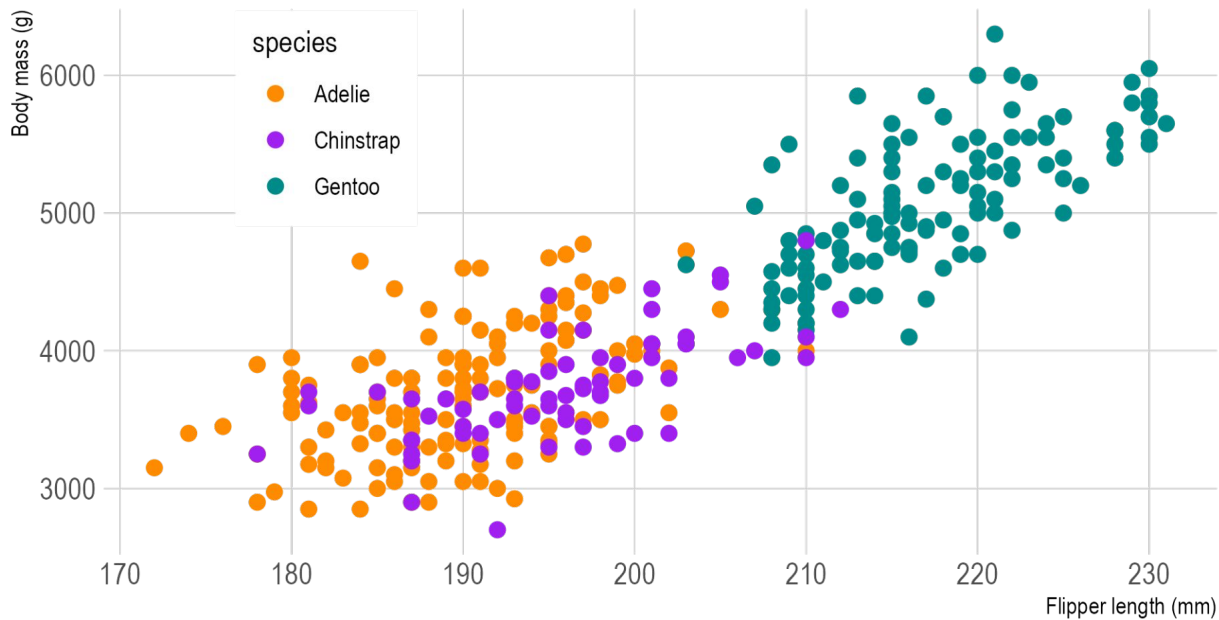
Ground living area partially explains sale price of apartments



Source: [From Data to Viz](#)

Penguin size, Palmer Station LTER

Flipper length and body mass for Adelie, Chinstrap and Gentoo Penguins



Source: [Data Visualization Guide](#)

Life Expectancy

● South America ● North America ● Asia ● Europe

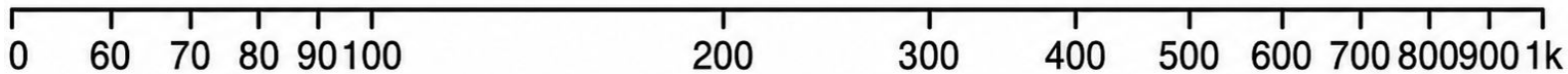
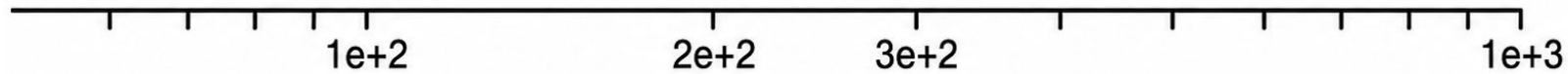
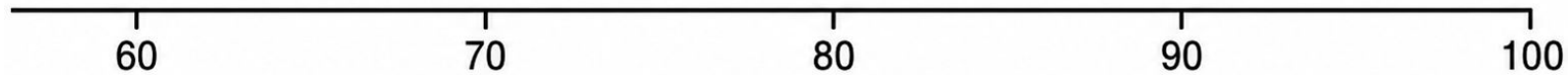
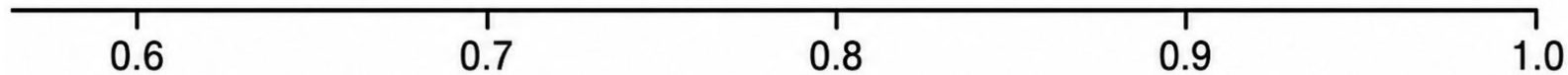
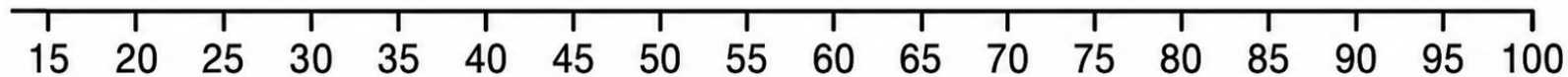


Source: [Ampler](#)

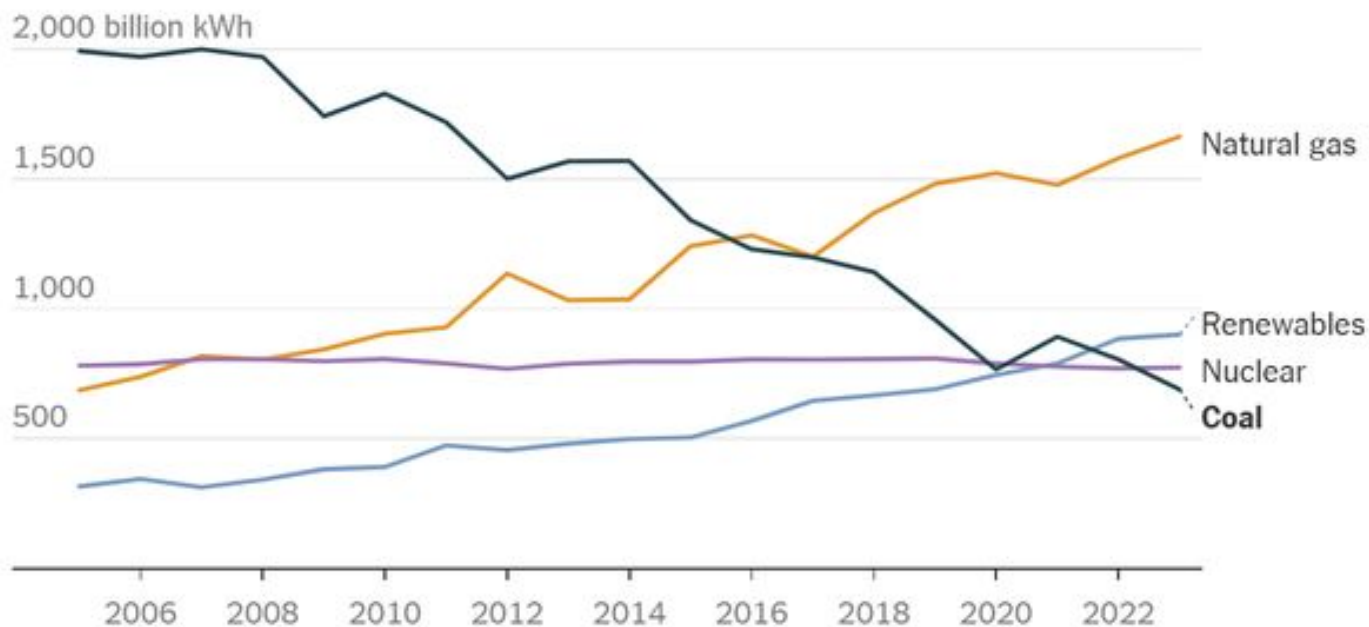
Scales

The translation between a data value and a visual value. The function that connects your table to your canvas.

- **Linear:** numbers map proportionally to position or length.
- **Logarithmic:** for data spanning orders of magnitude.
- **Ordinal:** categories map to positions or colors.
- **Color scales:** sequential, diverging, or categorical.



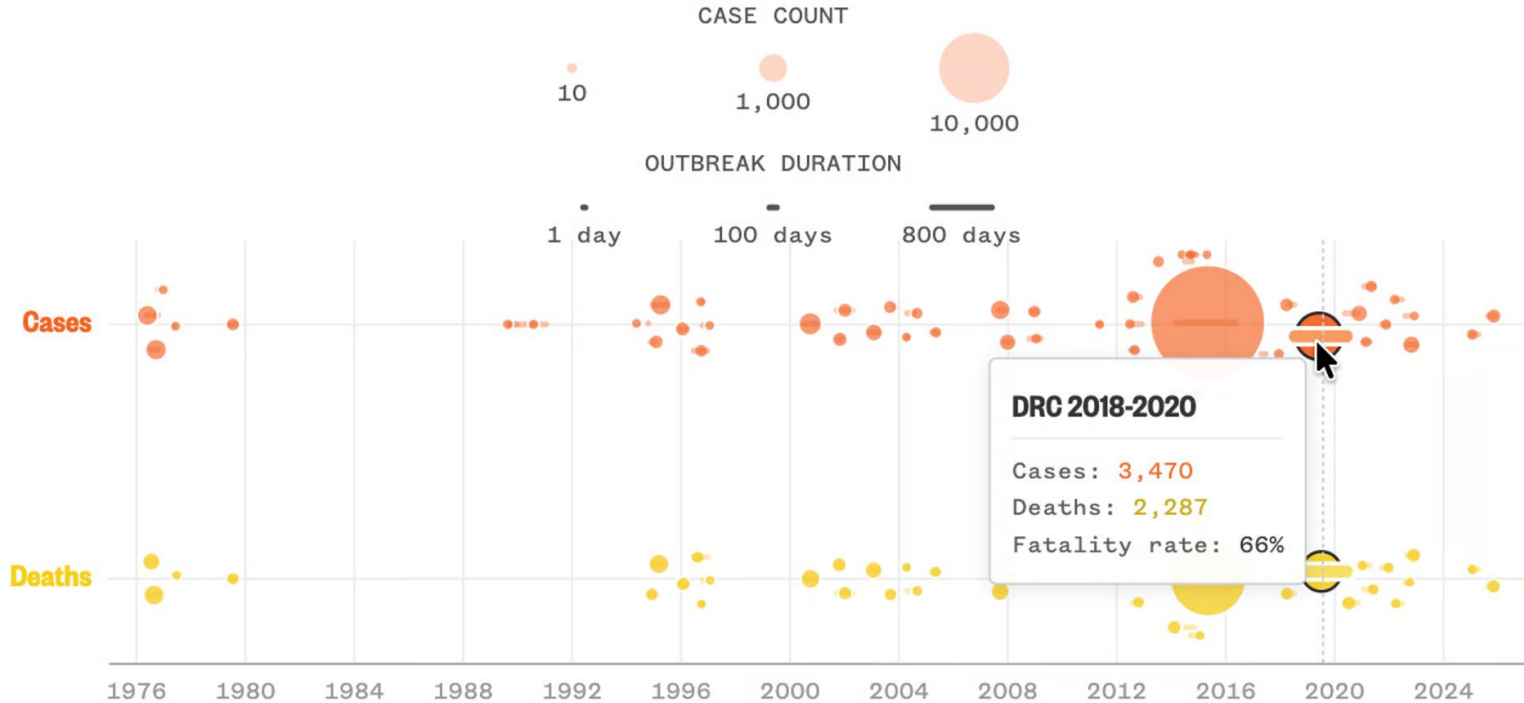
U.S. electricity generation by source



Note: Renewables include wind, solar, hydropower, geothermal and biomass. • Source: Rhodium Group • By The New York Times

Ebola is one of the world's deadliest viruses

The fatality rate of some types of Ebola can average around 50% to 60%. The Zaire virus, the most common species of Ebola, can kill 60% to 90% of people who get sick.



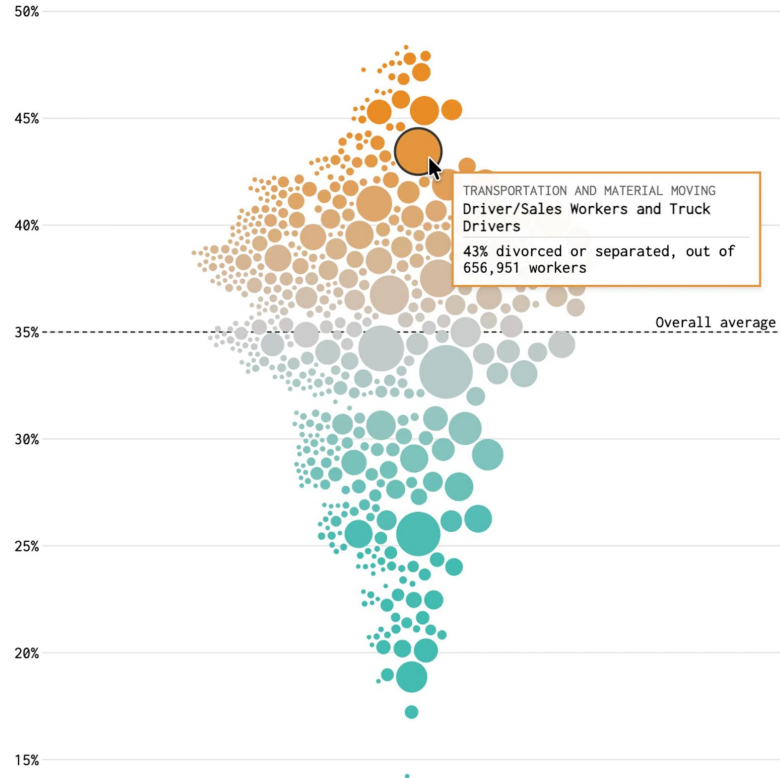
Source: U.S. Centers for Disease Control and Prevention

Graphic: Jiachuan Wu / NBC News

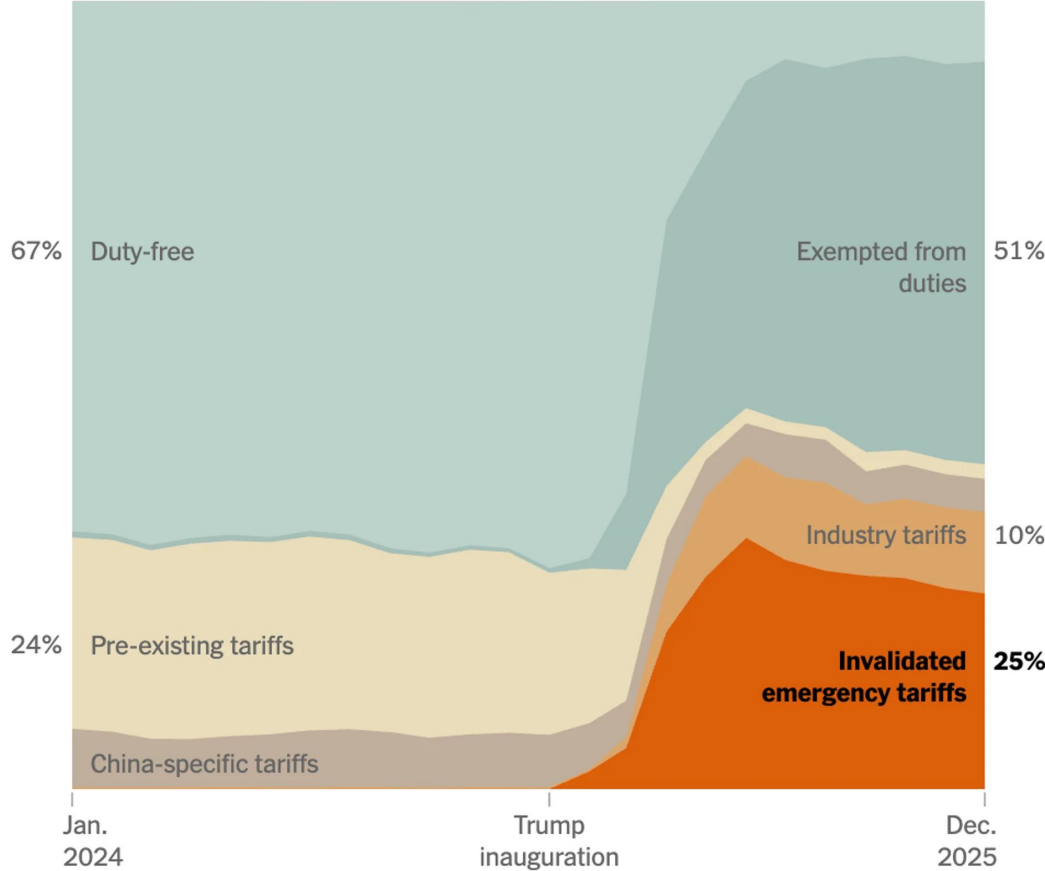
Divorce rate, by occupation

Search for or select an occupation below.

In 2024, among workers who married at least once, 35% were divorced or separated or had been divorced before.



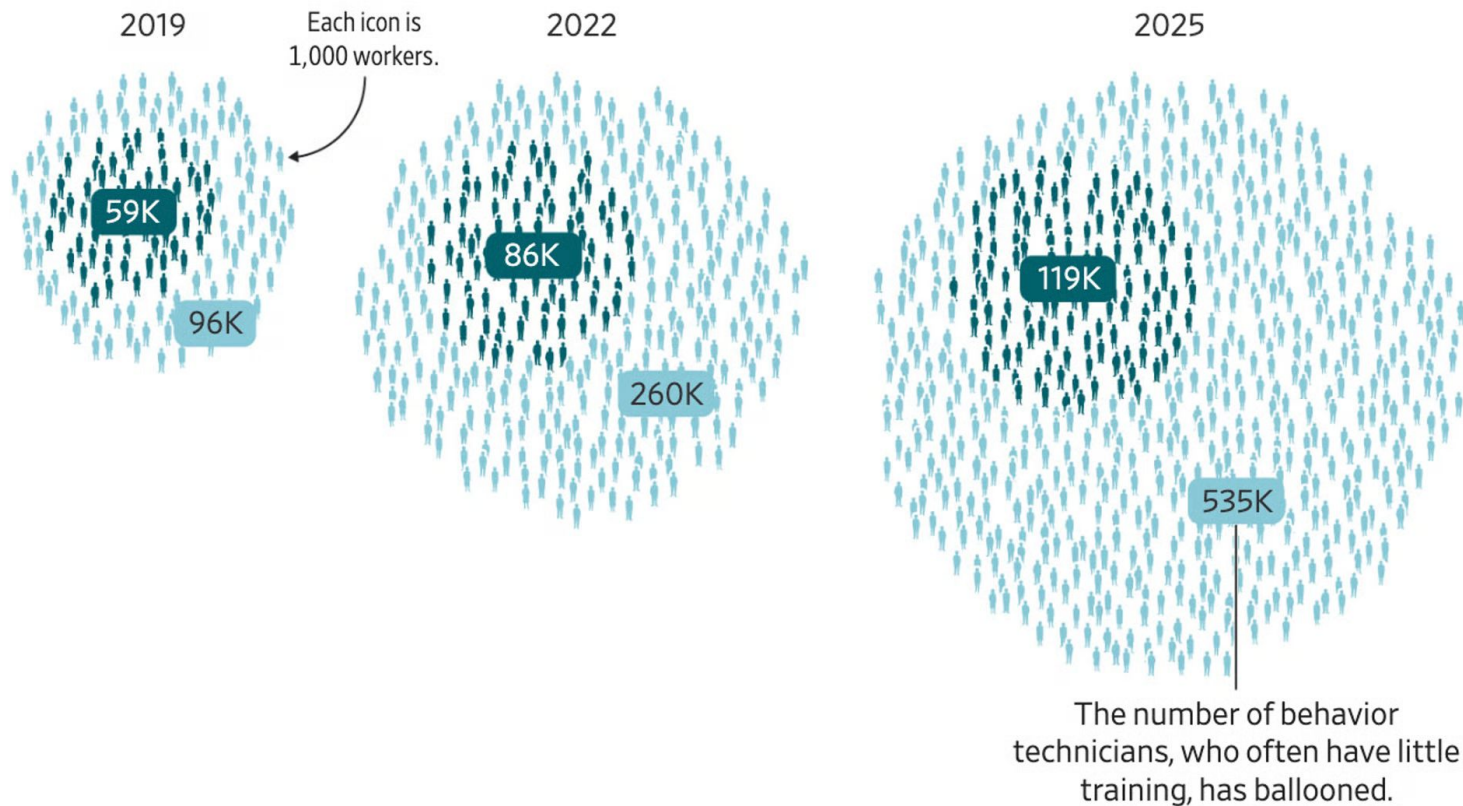
How the share of U.S. imports under different trade rules evolved



Source: New York Times analysis of U.S. Census Bureau international trade data.

Workers providing autism therapy*

Behavior analysts Behavior technicians



*Individual registrants in a federal provider database; not all workers register
Source: WSJ analysis of National Plan and Provider Enumeration System data

Andrew Mollica/WSJ

Some additional considerations

Truncated y-axis: starting a bar chart at something other than zero. Sometimes misleading, sometimes the only way to show meaningful variation. The difference matters.

Overloaded aesthetics: encoding the same variable twice (redundant encoding can actually help) vs. encoding too many variables at once (almost always hurts)

Chart junk: Tufte's concept of non-data ink. 3D charts, unnecessary gridlines, decorative elements. When does visual complexity serve the story and when does it obscure it?

The lie factor: when the visual representation of a change is proportionally larger or smaller than the actual change in the data.